

Understandable Autoformalization

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Introduction

Prediction

Once superhuman level of theorem proving and research in mathematics is achieved, recreational mathematics will become much more important than it is now.



Example 1: Suboptimal scenario

Shinichi Mochizuki may have proven the famous [ABC conjecture](#)

If true, this would be a breakthrough in number theory. Unfortunately, several years after the publication of the proof on the arXiv nobody else seems to fully understand his proof.

Question: How many people will be able to understand the proof by the time computers reach superhuman level at theorem proving? How many practical applications will the proof of the ABC conjecture have until then?

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Example 2: Optimal scenario

Fermat's [Sum of two squares Theorem](#) is the following result:

For every prime number p of the form $p = 4k + 1$, $k \in \mathbb{N}$, we have

$$p = m^2 + n^2 \text{ for some } m, n \in \mathbb{N}$$

The first proof of this theorem by Euler takes several pages.

In 1990 Don Zagier found an amazing “one-sentence proof”, see [\(Link\)](#).

Hundreds of millions of people should be able to understand Zagier's proof and there are probably millions who will find the proof fascinating and/or beautiful.

Remarks

There are hundreds (and maybe even thousands) of similar examples. Several dozens of them can be found in the book

Proofs from THE BOOK

by Martin Aigner and Günter M. Ziegler.

(These examples are from Number Theory, Geometry, Analysis, Combinatorics, and Graph Theory.)

More predictions

- Proof assistants will play a key role in mathematics in the next few years.
- Among proof assistants Lean will probably dominate but other proof assistants will fill suitable niches.
- Proof assistant focussing on readability can be very useful for recreational mathematics and the teaching of mathematics.

Interesting project

Use the latest LLMs and coding agents to find more “proofs from THE BOOK” by asking them to shorten standard proofs for important (and not too technical) theorems.

Once they are found (e.g. by using Lean) ask another coding agent to (re)formalize them using a proof assistant focussing on readability.

Autoformalization

Important use cases of Autoformalization

- In order to verify the correctness of a human proof, we could easily formalize it and apply a proof checker.
- Also currently available AI systems for finding proofs could formalize proofs on the fly (during proof search).
- In order to create a large dataset of proofs (of crucial importance for the development of AI systems for finding proofs), formalization must be automated.

Challenges

Math papers & text books present additional difficulties, not found in the standard “math problems” datasets which are usually “self-contained”:

- Apart from text they also contain diagrams & figures
- Gaps: obvious (or well-known) facts are often omitted in the proofs
- “Long distance” references/associations, requiring large context windows (including references between different papers/books)
- Changing and inconsistent notation (in particular, when different papers/books are involved)
- ...

Recent Breakthroughs in Autoformalization

Breakthrough for AF with customized agents (September 2025)

- **Gauss**, an autoformalization agent for Lean, was used to formalize the strong Prime Number Theorem (PNT) in Lean, addressing previously challenging analytic obstacles.
- Gauss produced approximately 25,000 lines of Lean code in three weeks, covering over 1,000 theorems and definitions.
- The Trinity environments on Morph Cloud enabled large-scale Lean verification with parallel support for many agents.

Breakthrough for AF with general coding agents (January 2026)

- Using a simple feedback loop between a coding agent and the [Megalodon](#) proof checker Josef Urban autoformalized a large portion of Munkres' textbook [Topology](#)
- Formalized theorems include major results such as Urysohn's lemma, Urysohn's metrization theorem, and the Tietze extension theorem.
- This suggests autoformalization may become cheap and widespread.
- [Link](#) to paper by Josef Urban.

Understandable Autoformalization

Goal

Develop proof assistants whose target language is more readable.

Important Example I

One of the first proof assistants with an emphasis on readability is the proof assistant [Isabelle](#) (with formal proof language [Isar](#)).

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Example for a proof in Isabelle

```
theorem sqrt2_not_rational:
  "sqrt 2  $\notin$   $\mathbb{Q}$ "
proof
  let ?x = "sqrt 2"
  assume "?x  $\in$   $\mathbb{Q}$ "
  then obtain m n :: nat where
    sqrt_rat: "?x = m / n" and lowest_terms: "coprime m n"
    by (rule Rats_abs_nat_div_natE)
  hence "m^2 = ?x^2 * n^2" by (auto simp add: power2_eq_square)
  hence eq: "m^2 = 2 * n^2" using of_nat_eq_iff power2_eq_square
  hence "2 dvd m^2" by simp
  hence "2 dvd m" by simp
  have "2 dvd n" proof -
    from <2 dvd m> obtain k where "m = 2 * k" ..
    with eq have "2 * n^2 = 2^2 * k^2" by simp
    hence "2 dvd n^2" by simp
    thus "2 dvd n" by simp
  qed
  with <2 dvd m> have "2 dvd gcd m n" by (rule gcd_greatest)
  with lowest_terms have "2 dvd 1" by simp
  thus False using odd_one by blast
qed
```

Example II: Naproche

The Naproche (NATural language PROOf CHEcking) family is a family of natural proof assistants using a Controlled Natural Language (CNL) instead of a formal proof language.

Historical background (Link)

- Evidence Algorithms and SAD (1960s–2008) by V. Glushkov and A. Paskevich
- Naproche 0 (2003–2015) by Peter Koepke and Bernhard Schröder
- Naproche-SAD (since 2017) by Peter Koepke et al.
- Naproche-ZF (aka “Felix”) (since 2022) by Adrian De Lon.

Example for a Naproche-ZF definition

Definition of Metric Space in Naproche-ZF

Example for a Naproche-ZF theorem & proof

Theorem 2 [Cantor]

There exists no surjection from A to $\mathcal{P}(A)$.

Proof

Suppose not.

Consider a surjection f from A to $\mathcal{P}(A)$.

Let $B = \{a \in A \mid a \notin f(a)\}$.

Then $B \in \mathcal{P}(A)$.

There exists $a' \in A$ such that $f(a') = B$ by [Definition 57](#).

Now $a' \in B$ iff $a' \notin f(a') = B$.

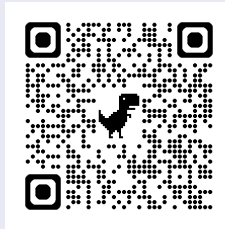
Contradiction.

Autoformalization with Naproche-ZF

Pilot project: textbook scale autoformalization in Naproche-ZF

Textbook: [Munkres' Topology Book](#)

Htmlized results: [Link](#)



GitHub repository with more resources: [Link](#)

Outlook

Applications of Understandable AF

Two natural candidates:

- 1 Recreational Math
- 2 Teaching of Math

Note

In some cases, like e.g. Isabelle, one can of course also expect further applications to research in math and software/hardware verification.

Example

Here is the formalization of [Zagier's proof](#) for Fermat's sum of two squares theorem with Naproche-ZF:

- Original formalized version: [Download HTML files](#)
- Shortened formalized version: [Download HTML files](#)

Remark

Zagier's involution has a beautiful geometric interpretation, see [Link](#)

Thank you!